

DiaTool-DPO

Multi-Turn DPO for Tool-Augmented LLMs

Enhancing dialogue flow via direct preference optimization with automatic paired trajectory construction.

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs trained only with Supervised Fine-Tuning (SFT) struggle with incomplete queries and out-of-scope requests.

1. Failing to ask follow-up questions for missing arguments.

2. Making hallucinated tool invocations.

3. Failing to reject inappropriate or out-of-scope tool calls.

Modeling TA-LLM Dialogue as a 5-State MDP

1. Initial State

No history. Out-of-scope requests return rejection.

2. Tool Selected w/o Complete Slots

Tool identified, required parameters missing. Slot-filling occurs.

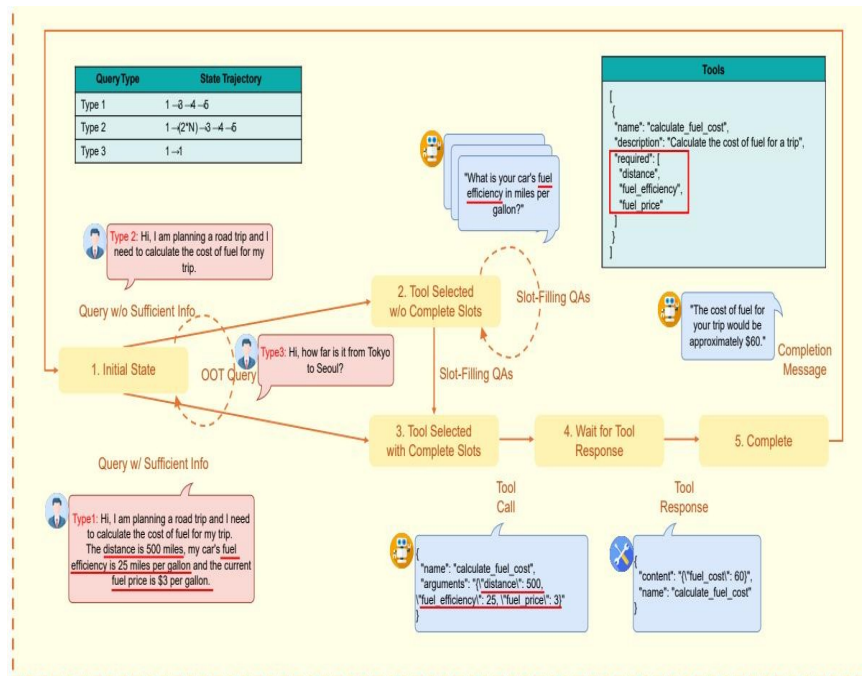
3. Tool Selected w/ Complete Slots

Selection and arguments determined.

4. Wait for Tool Response

Call executed, awaiting results.

5. Complete



Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query contains all required arguments. No slot-filling needed.
Type 2	$1 \rightarrow (2*N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks arguments. Slot-filling QA repeats N times.
Type 3	$1 \rightarrow 1$	Requested functionality unavailable. Tool call rejected.

Automatic Paired Trajectory Construction

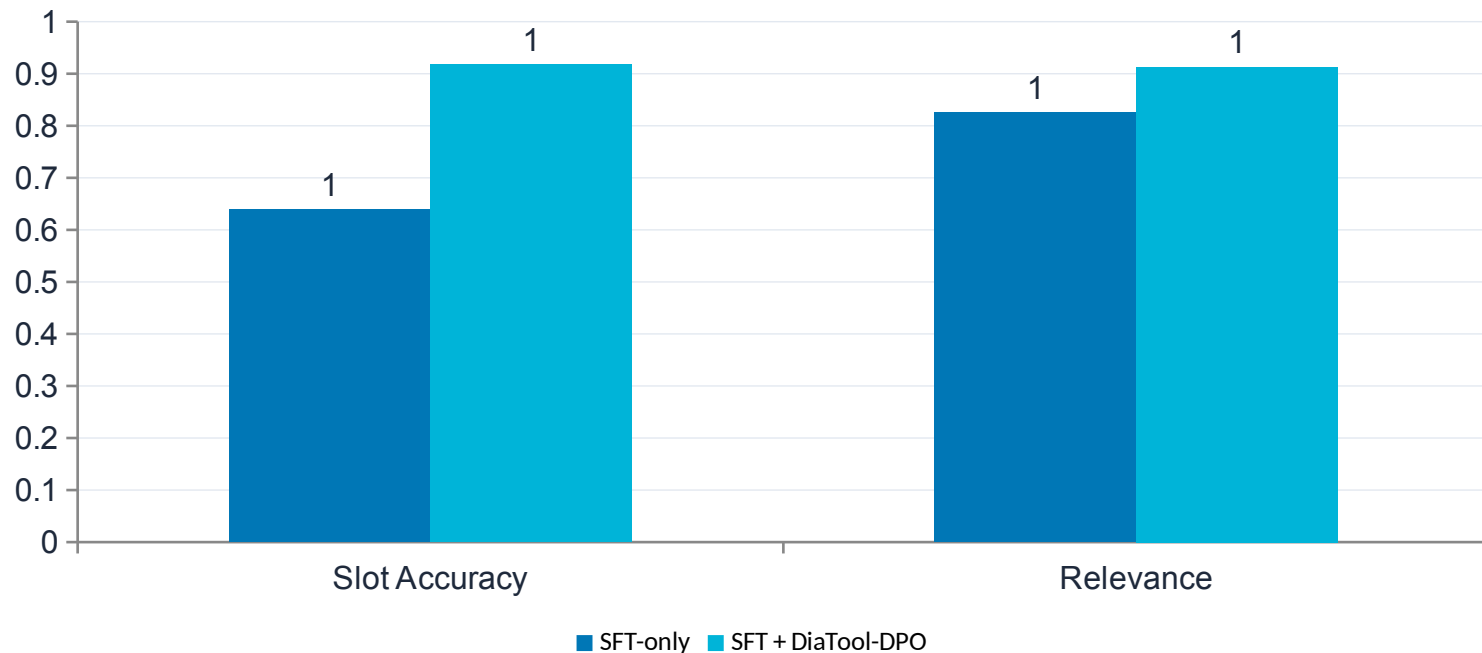
Dataset pairs chosen vs. rejected trajectories without human labels.

Type	Learning Lesson	Count	Difficulty
Type 1	Prevent redundant slot-filling	5,181	Easy, Hard
Type 1	Tool call accept	2,652	Easy, Hard
Type 2	Prevent slot hallucination	5,181	Easy, Hard
Type 2	Tool call accept	2,651	Easy, Hard
Type 3	Tool call reject	1,129	Hard

SFT + DiaTool-DPO Achieves Best Macro Average

Model	Method	Call	Comp.	Slot	Rel.	Micro	Macro
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Acc, +10.5% Rel. with DiaTool-DPO



Key Takeaways: RL-Based Dialogue Control

- MDP formulation enables systematic trajectory pairing without human annotation.
- DPO complements SFT for dialogue control; DiaTool-DPO consistently improves Macro Avg.
- Slot-filling and tool rejection see the largest gains (+43.5% and +10.5% on LLaMA-3-8B).
- The approach is language-agnostic, validating generalization.
- Opens path to GPT-4o-level conversational tool use on smaller open-source models.